The Substitution Rule
by: Dr.Hiba Omer
lec2

The Substitution Rule

- If $u=g(x)$ is differentiable function whose range is an interval I, and f is continuous on I, then

$$\int f(g(x))g'(x)dx = \int f(u)du$$

- The Substitution rule is proved using the Chain Rule for differentiation.
- The idea behind the substitution rule is to replace a relatively complicated integral by a simpler integral. This is accomplished by changing from the original variable x to a new variable u that is a function of x.
- The main challenge is to think of an appropriate substitution.

Example

$$\int \left(\underline{x+2}\right)^5 dx \qquad \text{Let } u = x+2$$

$$du = dx$$

$$\int u^5 du$$

$$\frac{1}{6}u^6 + C$$

$$\frac{\left(x+2\right)^6}{6} + C$$

<u>Don't</u> forget to substitute the value for u back into the problem!

Substitution rule examples

Example

To compute $\int e^{2x}\,dx$ set $u = 2x$.

Then $du = 2dx \Leftrightarrow dx = \frac{1}{2}du$ and

$$\int e^{2x}\,dx = \frac{1}{2}\int e^{u}\,du \;= \frac{1}{2}e^{u} + C$$

$$= \frac{1}{2}e^{2x} + C$$

We computed *du* by straightforward differentiation of the expression for *u*.

The substitution *u = 2x* was suggested by the function to be integrated. The main problem in integrating by substitution is to find the right substitution which simplifies the integral so that it can be computed by the table of basic integrals.

Example about choosing the substitution

Example Compute $\int \frac{\sqrt{x-1}}{x}dx.$

Here one is tempted to try the substitution $u = x-1,\ x = u+1,$ $dx = du.$ One gets $\int \frac{\sqrt{x-1}}{x}dx = \int \frac{\sqrt{u}}{u+1}du.$ This is not any easier.

Solution Substitute $u = \sqrt{x-1}.$ Then $x = 1+u^2,$

and $du = \frac{dx}{2\sqrt{x-1}} \Rightarrow dx = 2udu.$

This rewriting allows us to finish the computation using basic formulae.

$$\int \frac{\sqrt{x-1}}{x}dx = \int \frac{u}{1+u^2}2udu = 2\int\left(1-\frac{1}{1+u^2}\right)du = 2u - 2\arctan(u) + C$$

Next substitute back to the original variable.

Hence $\int \frac{\sqrt{x-1}}{x}dx = 2\sqrt{x-1} - 2\arctan(\sqrt{x-1}) + C$

The substitution rule for definite integrals

If *g'* is continuous on [*a*,*b*], and *f* is continuous on the range of *u=g(x)* then

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du$$

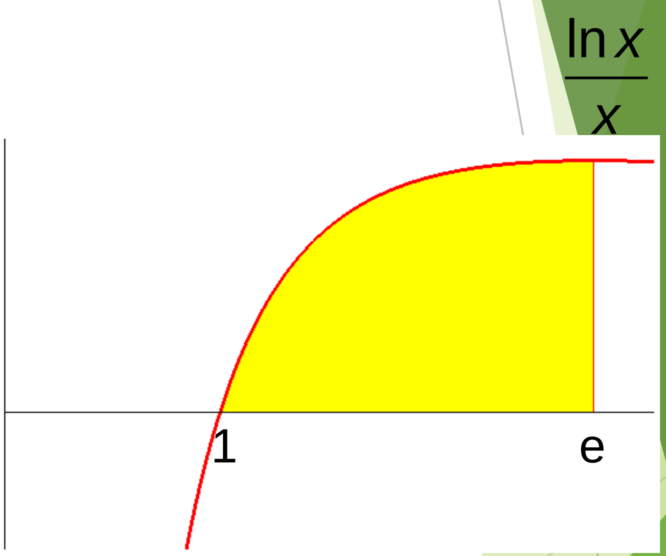
The substitution rule for definite integrals

Example

Compute $\int_1^e \frac{\ln x}{x}\,dx.$

The substitution $t = \ln x$ is suggested by the function to be integrated. We have $x = 1 \Rightarrow t = 0$ and $x = e \Rightarrow t = 1.$

Hence $\int_1^e \frac{\ln x}{x}\,dx = \int_0^1 t\,dt = \left.\frac{t^2}{2}\right]_0^1 = \frac{1}{2}.$



The area of the yellow domain is ½.

Example

$$\int_0^{\frac{\pi}{4}} \tan x \ \sec^2 x \ dx$$

Let $u = \tan x$

$$du = \sec^2 x \ dx$$

new limit

$$\int_0^1 u \ du$$

new limit

$$u(0) = \tan 0 = 0$$

$$u\left(\frac{\pi}{4}\right) = \tan\frac{\pi}{4} = 1$$

Find new limits

$$\frac{1}{2}u^2 \Big|_0^1$$

$$\frac{1}{2}$$

Example

$$\int_{-1}^{1} 3x^2\sqrt{x^3+1}\ dx$$

$$\text{Let } u = x^3+1 \qquad u(-1)=0$$

$$du = 3x^2\ dx \qquad u(1)=2$$

$$\int_0^2 u^{\frac{1}{2}}\ du$$

$$\left.\frac{2}{3}u^{\frac{3}{2}}\right|_0^2$$

Don’t forget to use the new limits.

$$\frac{2}{3}\cdot 2^{\frac{3}{2}} = \frac{2}{3}\cdot 2\sqrt{2} = \frac{4\sqrt{2}}{3}$$